

Uniform Convergence of $f_n(x) = x^n$

Proof with Complete Justifications

Problem Statement

Let $\langle f_n \rangle$ be a sequence of functions defined by $f_n(x) = x^n$ for $n \in \mathbb{N}$. Show that:

1. $\langle f_n \rangle$ is uniformly convergent on $[0, k]$ where $0 < k < 1$.
2. $\langle f_n \rangle$ is only pointwise convergent (not uniformly convergent) on $[0, 1]$.

Proof

Step 1: Pointwise Limit Function

We first determine the pointwise limit of $f_n(x) = x^n$ on $[0, 1]$.

- For $x \in [0, 1)$: Since $0 \leq x < 1$, we have $\lim_{n \rightarrow \infty} x^n = 0$.
- For $x = 1$: $\lim_{n \rightarrow \infty} 1^n = 1$.

Thus the pointwise limit function $f : [0, 1] \rightarrow \mathbb{R}$ is:

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Step 2: Uniform Convergence on $[0, k]$ where $0 < k < 1$

We apply the **Supremum (M_n) Test**: a sequence $\langle f_n \rangle$ converges uniformly to f on a set S if and only if

$$M_n := \sup_{x \in S} |f_n(x) - f(x)| \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since $x \in [0, k]$ with $k < 1$, the pointwise limit is $f(x) = 0$ throughout $[0, k]$. Therefore:

$$M_n = \sup_{x \in [0, k]} |x^n - 0| = \sup_{x \in [0, k]} x^n.$$

Claim: $f_n(x) = x^n$ is monotonically increasing on $[0, k]$.

Justification. Differentiating, $f'_n(x) = nx^{n-1}$. For $x \in [0, k]$ with $k > 0$ and $n \geq 1$, we have $x \geq 0$, so $f'_n(x) = nx^{n-1} \geq 0$. Hence f_n is non-decreasing on $[0, k]$, and strictly increasing on $(0, k]$. $\square$

Since f_n is increasing on $[0, k]$, its supremum is attained at the right endpoint $x = k$:

$$M_n = k^n.$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} M_n = \lim_{n \rightarrow \infty} k^n = 0, \quad \text{since } 0 < k < 1.$$

By the M_n -test, $\langle f_n \rangle$ **converges uniformly** to $f(x) = 0$ on $[0, k]$. □

Step 3: Only Pointwise Convergence on $[0, 1]$

We now show that the convergence *fails* to be uniform on the full interval $[0, 1]$. We use the contrapositive of the following classical result.

Theorem 1 (Continuity of Uniform Limits). *Let $\langle f_n \rangle$ be a sequence of functions, each continuous on an interval $I \subseteq \mathbb{R}$. If $f_n \rightarrow f$ uniformly on I , then f is continuous on I .*

We apply the **contrapositive**: if the limit function f is *discontinuous* on I , then the convergence cannot be uniform on I .

Proof of Non-Uniformity on $[0, 1]$ Each $f_n(x) = x^n$ is a polynomial, hence continuous on $[0, 1]$ for every $n \in \mathbb{N}$.

However, the pointwise limit f established in Step 1 is *discontinuous* at $x = 1$, since:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 0 = 0 \neq 1 = f(1).$$

By the contrapositive of Theorem 1, a sequence of continuous functions cannot converge uniformly to a discontinuous limit. Therefore, $\langle f_n \rangle$ does **not** converge uniformly on $[0, 1]$.

Hence $\langle f_n \rangle$ is only **pointwise convergent** on $[0, 1]$. □

Remark. The non-uniform convergence can also be verified directly: choose $x_n = k^{1/n}$ for any fixed $0 < k < 1$, so $x_n \in [0, 1)$ and $x_n \rightarrow 1$. Then $f_n(x_n) = (k^{1/n})^n = k \not\rightarrow 0 = f(x_n)$, confirming that no single N works for all x near 1.